GARVATA: PRECISION TECHNOLOGY AIRFLOW

EFF: >95%
PWR: 1.75kW

SMART AHU & EC FAN SOLUTIONS

AI-driven optimization for energy-efficient, autonomous AHU operations achieving 50-55% energy reduction.

AI_MODE: ACTIVE
SFP: 0.31 W/(m³/s)

Challenges in AHU Operations

PROBLEM STATEMENT

Operational Hurdles

LEVEL I

Energy Targets Unmet

Current systems struggle to achieve significant energy reduction goals due to fixed-speed operations.

Efficiency Gaps

Manual interventions lead to suboptimal fan speeds and wasted operational capacity.

Air Quality Variance

Fluctuating Temperature, RH, and CO2 levels impact occupant comfort and compliance.

Reactive Maintenance

Preventive maintenance is often delayed, leading to unexpected downtime and repairs.

Systemic Integration

LEVEL II

Automation Complexity

High implementation costs, dependency on silos, and lack of offline operation capabilities.

BMS Integration

Existing solutions often struggle with seamless data exchange and control integration with Building Management Systems.

Fire System Safety

Gaps in automated response protocols during fire emergencies due to poor system handshakes.

Solution Overview

Transforming HVAC operations through intelligence and autonomy

CORE VALUE PROPOSITION

Energy Saving

50-55%

Significant reduction in power consumption through optimized EC fan modulation.

AI-Driven Control

Zero

Manual intervention required. Closed-loop AI algorithms manage all variables.

Autonomous Ops

100%

Fully autonomous AHU operation adapting to real-time environmental demands.

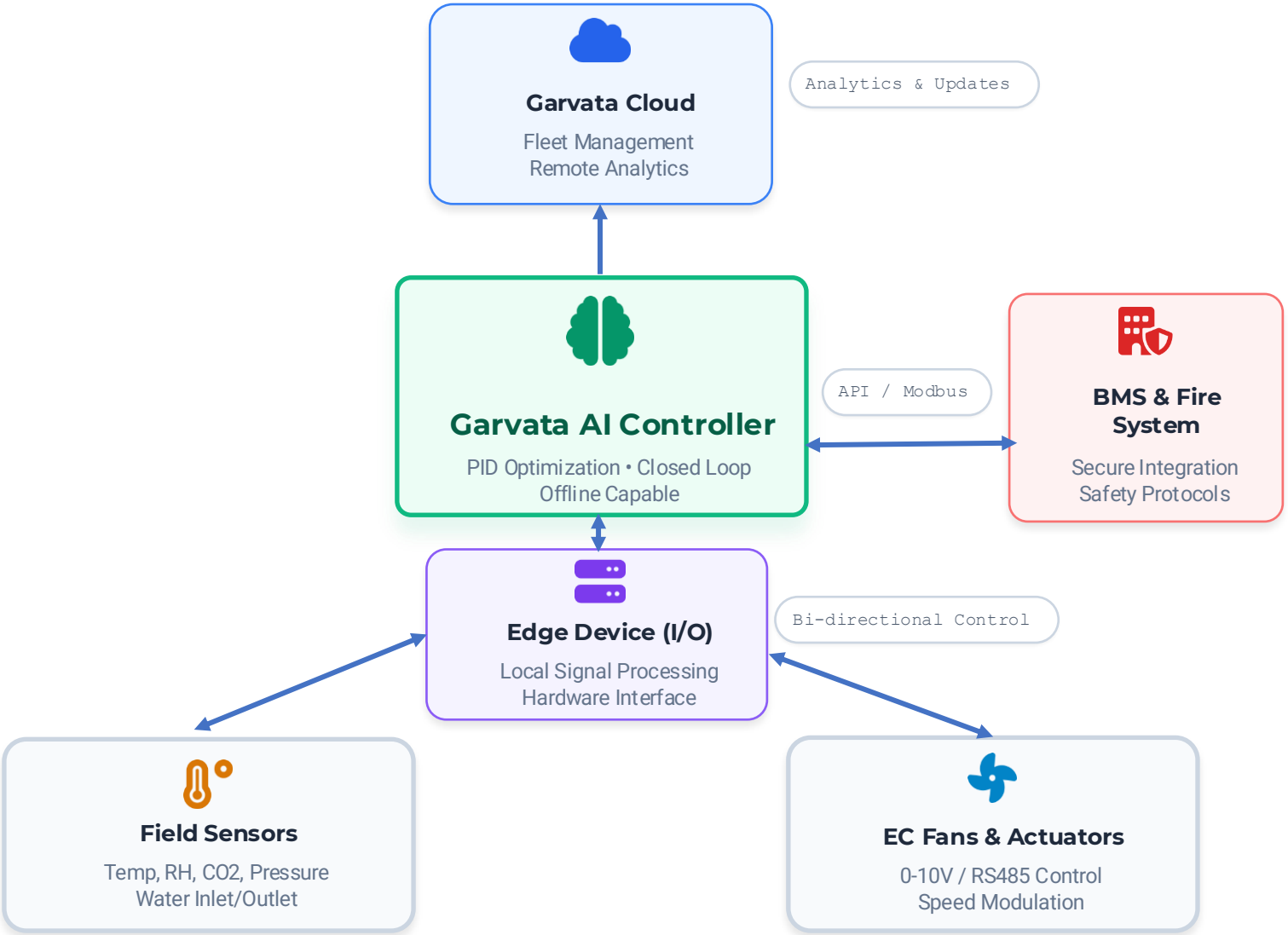
Extended Capabilities

Beyond basic control

- Real-time Optimization
  Temp, RH, CO2, Pressure
- Seamless Integration
  BMS & Fire via APIs
- Central Supervision
  Remote monitoring & override

System Architecture

End-to-end data flow from field sensors to cloud analytics



Operating Principle

AI-Optimized Control Logic & Execution Sequence

AI-Optimized PID Core

Multi-variable tuning algorithm adapting to Temperature, RH, CO_2, and Pressure dynamics simultaneously.

Execution Cadence

Control Loop: Every 1 minute

Fan Adjustment: Every ~5 minutes window (configurable) to prevent hunting and wear.

Industry Modes

Pharma/R&D: Priority on Pressure (Zero Recirculation).

Comfort: Priority on Temp/RH efficiency.

Safety Limits Always Active

EC Fan Technical Specifications

High-Efficiency Electronically Commutated Fan Data

>95%
MOTOR EFFICIENCY (IE5)

1.75kW
RATED POWER / FAN

62-70dB
LOW NOISE OPERATION

Retrofit Ready

Compact form factor allows seamless replacement of existing induction motors with minimal duct modification.

Detailed Parameters

SERIES: EC-GEN-X

AERODYNAMICS & POWER

Airflow Range	12,500 - 19,000 m³/h
Static Pressure	Up to 750 Pa
Specific Fan Power (SFP)	0.31 - 0.36 W/(m³/s)
Impeller Material	Aluminum Alloy

CONTROL & INTERFACE

Communication	Modbus RTU / RS485
Analog Input	0-10V DC / 4-20mA
Protection Class	IP55 (Dust/Water)
Power Supply	3-Phase ~380-480V

- IEEE 519 Compliant (Low Harmonics)
- CE Certified
- RoHS Compliant

Key Benefits Summary

Delivering value through efficiency, intelligence, and integration

50-55% Energy Savings

Proven reduction in AHU power consumption through optimized EC fan modulation and AI logic.

Autonomous Operation

Full autonomous control reduces human dependency and eliminates manual errors.

Stable Air Quality

Precise regulation of Temp, RH, CO2 and Differential Pressure for occupant comfort.

Predictive Insights

Continuous monitoring of filter ΔP and motor health enables proactive maintenance.

Seamless Integration

Robust API and Modbus connectivity for existing BMS and critical Fire Safety systems.

Low Harmonics

IEEE 519 compliant EC motors protect power quality and extend equipment lifespan.

Modular Retrofit

Compact form factor allows for phased deployment with minimal downtime.

System Resilience

Edge devices support offline local control and manual overrides for maximum reliability.

Performance Case Study

Field Validation: Before vs After Transformation

POWER CONSUMPTION

2.9 kW

6.8 kW

↓ 57% Saved

CFM DELIVERY

22,180

18,345

↑ 21% Increase

EFFICIENCY (KW/TR)

0.10

0.31

↓ 72% Improved

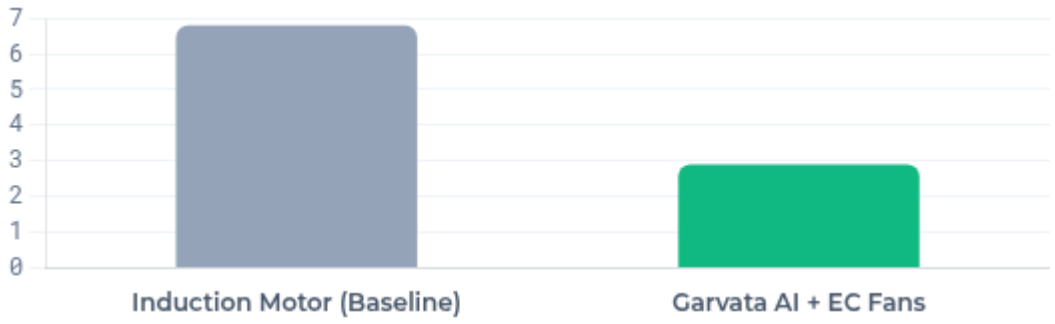
SPEC. POWER (KW/CFM)

-64%

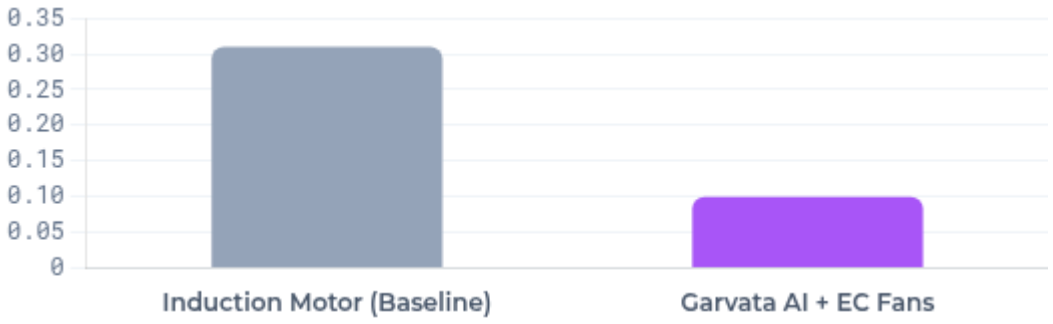
Relative

↓ Massive Gain

Power Consumption (kW)



Specific Energy (kW/TR)



ROI & Payback Analysis

Financial justification and long-term value assessment

ESTIMATED PAYBACK PERIOD

19 Months

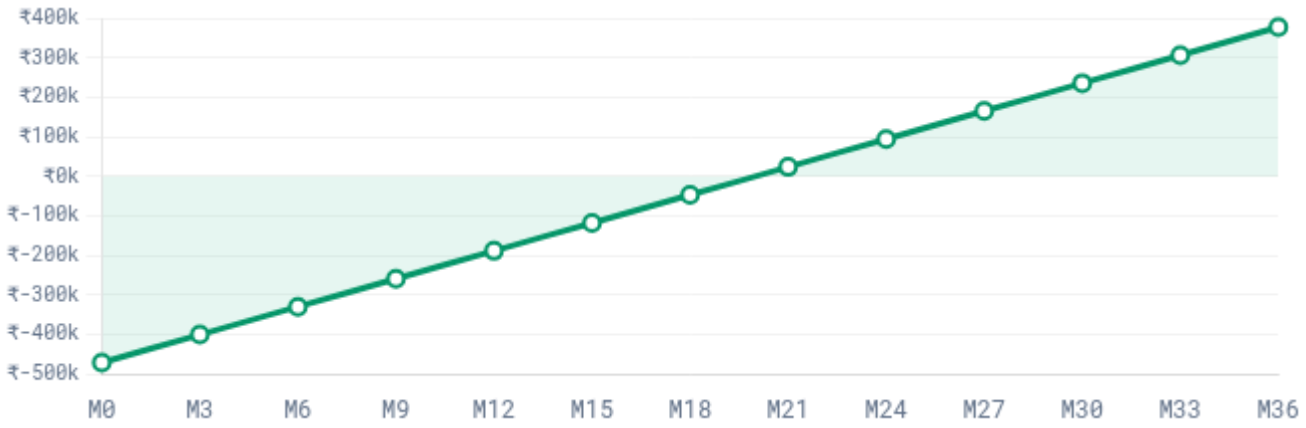
Based on current usage patterns

DAILY OPERATIONAL SAVINGS

INR 785/AHU

Immediate Impact

Cumulative Savings Projection (3 Years)



Maintenance Reduction

Predictive alerts reduce breakdown frequency and eliminate unnecessary scheduled checks.

Compliance Support

Automated logging and precise pressure control ensure adherence to Pharma/R&D standards.

Assumptions

Includes hardware, installation & commissioning costs. Based on 24x7 operation.

Ready to Transform?

Join us in revolutionizing HVAC efficiency

NEXT STEPS

PILOT DEPLOYMENT PLAN

Identify Pilot AHUs

Select 2-3 critical units with high operating hours for maximum impact demonstration.

Baseline Measurement

1-week data logging of existing induction motor performance to establish benchmark.

Deployment

Installation of Edge Devices and EC Fans (Retrofit typically completed in <8 hours).

Go-Live & Optimization

Activate AI Controller. Immediate realization of 50%+ energy savings.

Thank You!

Schedule your site assessment or request a live demo today.

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Book Assessment